

ABIGAIL LEWIS, Ph.D.

Smithsonian Environmental Research Center, Edgewater, MD, USA
 abigail.sl.lewis@gmail.com | (262) 565-7269

EDUCATION

- 2019–2024 **Ph.D. in Biological Sciences**, Virginia Tech
- Dissertation: *Oxygen dynamics in the bottom-waters of lakes: Understanding the past to predict the future*; Advisor: Dr. Cayelan Carey
 - Preparing the Future Professoriate Certificate, Associate of the Graduate Academy for Teaching Excellence
- 2015–2019 **B.A. in Biology**, Spanish minor, Pomona College
- Thesis: *Effects of artificial light at night on benthic freshwater carbon cycling*; Advisor: Dr. Wallace Meyer III
 - *Magna cum laude*, Phi Beta Kappa, Distinction in the Senior Exercise

RESEARCH EXPERIENCE

- 2024–present **Postdoctoral Climate Change Fellow**, Smithsonian Institution
- 2019–2024 **NSF Graduate Research Fellow**, Virginia Tech
Institute for Critical Technology and Applied Science (ICTAS) Doctoral Scholar, Virginia Tech
- 2018 **NSF REU Intern**, University of Wisconsin Madison
- 2017 **NSF REU Intern**, Cary Institute of Ecosystem Studies (CIES)
- 2016–2017 **Research Assistant**, Pomona College

FELLOWSHIPS (total of ~\$579,120)

- 2024 **Fulbright Postdoctoral Fellowship** (Spain; ~\$32,500, *declined*)
Science Communication In the Parks Fellowship, Ecological Society of America and the U.S. National Parks Service (\$3700)
- 2023 **Smithsonian Climate Change Fellowship** (\$133,920)
- 2019 **NSF Graduate Research Fellowship** (\$141,000)
ICTAS Doctoral Fellowship, Virginia Tech (~\$192,000)
Cunningham Fellowship, Virginia Tech (~\$76,000; *declined*)
Interfaces of Global Change Fellowship, Virginia Tech

SELECTED GRANTS, AWARDS, AND HONORS (total of \$28,000)

- 2024 **Outstanding Doctoral Student Award**, Virginia Tech College of Science (\$500)

Trailblazing Award in Community & Outreach, Ecological Society of America Student Section (\$700)

Outstanding Student Presentation Award, American Geophysical Union (\$200)

2023 **Noel Krieg Graduate Fellowship**, Virginia Tech (\$2,000)

2022 **Make-a-Difference Award**, Virginia Tech College of Science Roundtable (\$10,000)

Mary and George Schaeffer Stream Team Excellence Award, Virginia Tech (\$1,000)

Katherine S. McCarter Graduate Student Policy Award, Ecological Society of America

Graduate Academy for Teaching Excellence Associate, Virginia Tech

2021 **AGU Sharing Science Grant** (\$1,000)

- Funded the development of a flipped science fair, where elementary students judged graduate student science fair posters

2019 **John Stauffer Scholarship for Academic Merit**, Pomona College (\$12,000)

- Description: "An award is made to a senior student majoring in Physics, Chemistry, or a related science, who exhibits the highest academic promise"

Vaile Prize, Pomona College (\$600)

Distinction in the senior exercise, Pomona College

PEER-REVIEWED PUBLICATIONS (* indicates mentored undergraduate student). *Total of 32 published or in review; 11 first-author*

In review **Lewis, A. S. L.**, J. R. Holmquist, A. N. Al-Haj, Z. N. Read, R. L. Rich, J. P. Megonigal, and G. L. Noyce. Warming decreases the predictability of coastal wetland methane emissions. In review at *Ecosphere*.

Lewis, A. S. L., D. C. Richardson, D. W. Howard, C. C. Carey, Y. Amitai, S. Bansal, E. de Eyto, H-P Grossart, K. K. Hoffman, R. Hovel, L. Knoll, B. M. Kraemer, I. Oleksy, M. Schmid, R. Schwefel, D. Straile, X. Sun, G. A. Weyhenmeyer, W. M. Woelmer, S. Wollrab, P. Znachor. How should we analyze seasons and seasonality in lakes? In revision at *Limnology and Oceanography Letters*.

Hounshell, A. G., **A. S. L. Lewis**, D. W. Howard, H. L. Wander, M. E. Lofton, P. C. Hanson, and C. C. Carey. Dissolved organic carbon dynamics are driven by water temperature, primary production, and anoxia over five years of whole-ecosystem experiments in a eutrophic reservoir. In revision at *Aquatic Sciences*.

- Colton, M., T. Fanara, S. Babin, **A. S. L. Lewis**, M. Dietze, C. Brown, M. Cotnoir, A. Caldas, K. Boyd. Ecological Forecasting for Readiness, Resilience, and Response. In revision at *Bulletin of the American Meteorological Society*.
- Rabaey, J. S., **A. S. L. Lewis**, K. Attermeyer, P. Aurich, S. Bansal, M. Bartosiewicz, B. L. Bertolet, I. Bussmann, S. B. Cadieux, E. Calamita, C. Capelli, C. C. Carey, C. Cillero, F. Clayer, S. L. D'Ambrosio, T. A. Davidson, B. R. Deemer, B. A. Denfeld, W. Eckert, C. Esposito, P. Ford, A. Gorsky, N. A. Griffiths, H-P. Grossart, D. P. Hamilton, M. A. Holgerson, B. J. Huser, T. Iwata, J. Jansen, S. Jones, S. Juutinen, P. Kortelainen, M. Koschorreck, T. Kragh, A. Laas, T. Larmola, S. Läubli, I. Laurion, M. F. Lehmann, L. Liu, P. J. Martikainen, A. Matoušů, S. A. McCord, J. J. Montes-Pérez, D. Nizzoli, C. Ordóñez, M. Peacock, R. M. Pilla, V. Preskienis, J. Pu, T. Riis, T. Saarela, A. B. Santoso, C. J. Schubert, A. Sepulveda-Jauregui, B. S. Sherman, J. S. Sør, K. J. Stenehjøm, K. J. Stenehjøm, K. E. D. Strock, K. Tsuchiya, K. Wendt-Potthoff, G. A. Weyhenmeyer, P. Znachor, and J. Zopfi. Carbon dioxide and methane depth-profile concentrations in 544 lakes, ponds, and reservoirs worldwide. In revision at *Scientific Data*.
- Holmquist, J. R., A. R. Carr, and **A. S. L. Lewis**. Continuous Salinity Mapping for Coastal Wetlands in the United States. In revision at *Remote Sensing of the Environment*.
- Sjodin, A., G. Stokes, C. Gallagher, M. Javadian, **A. S. L. Lewis**, C. Schreiner, L. Super, B. Tuschhoff, A. Willson, M. C. Dietze, J. MacLachlan, J. Peters. Toward a standard lexicon for ecological forecasting. In revision at *Frontiers in Ecology and the Environment*.
- 2025 **Lewis, A. S. L.**, D. Mercado-Bettin, and C. C. Carey. 2025. Ecological memory contributes to a spatial mismatch in water quality trends between the surface and bottom waters of 382 temperate lakes. *Ecology Letters*, 28(11): e70243. <https://doi.org/10.1111/ele.70243>
- Olsson, F. E., C. C. Carey, G. Harrison, R. Ladwig, M. F. Lapeyrolerie, **A. S. L. Lewis**, M. E. Lofton, F. Montealegre-Mora, J. S. Rabaey, C. J. Robbins, X. Yang, R. Q. Thomas. 2025. What can we learn from 100,000 freshwater forecasts? A synthesis from the NEON Ecological Forecasting Challenge. *Ecological Applications*, 35(1): e70004. <https://doi.org/10.1002/eap.70004>
- Wood, C. E., M. E. Schreiber, C. E. Bauer, A. Breef-Pilz, **A. S. L. Lewis**, N. W. Hammond, R. P. McClure, M. F. Verne, C. C. Carey. 2025. Preemptive activation of hypolimnetic oxygenation improves control of Fe and Mn in drinking water reservoirs. *Water Research*, 124722. <https://doi.org/10.1016/j.watres.2025.124722>
- 2024 **Lewis, A. S. L.**, A. Breef-Pilz, D. W. Howard, M. E. Lofton, F. Olsson, H. L. Wander, C. E. Wood, M. E. Schreiber, C. C. Carey. 2024. Reservoir Drawdown Highlights the Emergent Effects of Water Level Change on Reservoir Physics, Chemistry, and Biology. *JGR Biogeosciences*, 129, e2023JG007780. <https://doi.org/10.1029/2023JG007780>

- Lewis, A. S. L.**, M. P. Lau, S. F. Jane, K. C. Rose, Y. Be'eri-Shlevin, S. H. Burnet, F. Clayer, H. Feuchtmayr, H. P. Grossart, D. W. Howard, H. Mariash, J. Delgado Martin, R. L. North, I. Oleksy, R. M. Pilla, A. P. Smagula, R. Sommaruga, S. E. Steiner, P. Verburg, D. Wain, G. A. Weyhenmeyer, C. C. Carey. 2024. Anoxia Begets Anoxia: a positive feedback to the deoxygenation of temperate lakes. *Global Change Biology*, 30(1), e17046. <http://doi.org/10.1111/gcb.17046>
- Reported in 36 news stories from 34 outlets ([Altmetric](#))
- Mesman, J. P., C. C. Barbosa, **A. S. L. Lewis**, F. E. Olsson, S. Calhoun-Grosch, H.-P. Grossart, R. Ladwig, R. S. La Fuente, K. Münzner, L. G. T. Nkwale, R. M. Pilla, K. Suresh and D. J. Wain. 2024. Challenges of Open Data in aquatic sciences: issues faced by data users and data providers. *Frontiers in Environmental Science*, 12:1497105. <https://doi.org/10.3389/fenvs.2024.1497105>
- Howard, D. W., J. A. Brentrup, D. C. Richardson, **A. S. L. Lewis**, F. E. Olsson, and C. C. Carey. Variability in ice cover does not affect annual metabolism estimates in a small eutrophic reservoir. *JGR Biogeosciences*, 129, e2024JG008057. <https://doi.org/10.1029/2024JG008057>
- Malmborg, C. A., A. M. Willson, L. M. Bradley, M. A. Beatty, D. H. Klings, G. Koren, **A. S. L. Lewis**, K. Oshinubi, and W. M. Woelmer. Defining Model Complexity: An Ecological Perspective. *Meteorological Applications*, 31(3), e2202. <https://doi.org/10.1002/met.2202>
- Wander, H. L., **A. S. L. Lewis**, D. W. Howard, M. E. Lofton, W. M. Woelmer, B. L. Brown, and C. C. Carey. 2024. Zooplankton community structure and diel migration patterns vary over hours, days, and years in the pelagic and littoral zone of a eutrophic reservoir. *Journal of Plankton Research*, fbae017. <https://doi.org/10.1093/plankt/fbae017>
- Man, X., C. Lei, K. A. Bierlein, L. D. Bryant, **A. S. L. Lewis**, C. C. Carey, J. C. Little. 2024. Computationally characterizing the diffusive boundary layer in lakes and reservoirs. *Journal of Soils and Sediments*, 24, 2132–2143. <https://doi.org/10.1007/s11368-024-03767-0>
- 2023 **Lewis, A. S. L.**, C. R. Rollinson, A. J. Allyn, J. Ashander, S. Brodie, C. B. Brookson, E. Collins, M. C. Dietze, A. S. Gallinat, N. Juvigny-Khenafou, G. Koren, D. J. McGlinn, J. A. Peters, N. R. Record, C. J. Robbins, J. Tonkin, and G. M. Wardle. 2023. The power of forecasts to advance ecological theory. *Methods in Ecology and Evolution*, 14(3), 746–756. <https://doi.org/10.1111/2041-210X.13955>
- Lewis, A. S. L.**, M. E. Schreiber, B. R. Niederlehner, A. Das*, N. W. Hammond, M. E. Lofton, H. L. Wander, C. C. Carey. 2023. Effects of Hypoxia on Coupled Iron and Carbon Cycling Differ by Timescale in Two Freshwater Reservoirs. *JGR: Biogeosciences*, 128(1), e2022JG007071. <https://doi.org/10.1029/2022JG007071>
- Lewis, A. S. L.**, D. W. Howard, G. Koren, C. Kowalski, J. McLachlan, J. A. Peters, O. Tabares, G. Smies. 2023. Ethics in ecological forecasting: a case-based learning set. *Teaching Issues and Experiments in Ecology*, 19(13). <https://doi.org/10.25334/5D99-Y019>

- Lewis, A. S. L.**, G. O'Malley, G. K. Palissery, A. Hensley, C. López Lloreda, C. Perez, E. K. Bueren. 2023. Flipped Science Fair invites children to judge graduate student posters through a university-community partnership. *Journal of STEM Outreach*, 6(1), 1–12. <https://doi.org/10.15695/jstem/v6i1.14>
- Thomas, R.Q., C. Boettiger, C.C. Carey, M.C. Dietze, L.R. Johnson, M.A. Kenney, J.S. Mclachlan, J.A. Peters, E.R. Sokol, J.F. Weltzin, A. Willson, W.M. Woelmer, and **Challenge Contributors**. 2023. The NEON Ecological Forecasting Challenge. *Frontiers in Ecology and the Environment*, 21(3), 112–113. <https://doi.org/10.1002/fee.2616>
- Woelmer, W. M., A. G. Hounshell, M. E. Lofton, H. L. Wander, **A. S. L. Lewis**, D. Scott, C. C. Carey. 2023. The importance of time and space in biogeochemical heterogeneity and processing along the reservoir ecosystem continuum. *Aquatic Sciences*, 85(2), 66. <https://doi.org/10.1007/s00027-023-00959-7>
- 2022 **Lewis, A. S. L.**, W. M. Woelmer, H. L. Wander, D. W. Howard, J. W. Smith, R. P. McClure, M. E. Lofton, N. W. Hammond, R. S. Corrigan, R. Q. Thomas, C. C. Carey. 2022. Increased adoption of best practices in ecological forecasting enables comparisons of forecastability across systems. *Ecological Applications*, 32(2), e02500. <https://doi.org/10.1002/eap.2500>
- Jane, S., J. L. Mincer, M. P. Lau, **A. S. L. Lewis**, J. T. Stetler, K. C. Rose. 2022. Longer duration of seasonal stratification contributes to widespread increases in lake hypoxia and anoxia. *Global Change Biology*, 29(4), 1009–1023. <https://doi.org/10.1111/gcb.16525>
- Lofton, M. E., D. W. Howard, R. P. McClure, H. L. Wander, W. M. Woelmer, A. G. Hounshell, **A. S. L. Lewis**, C. C. Carey. 2022. Experimental thermocline deepening alters vertical distribution and community structure of phytoplankton in a four-year whole-reservoir manipulation. *Freshwater Biology*, 67(11), 1903–1924. <https://doi.org/10.1111/fwb.13983>
- Carey C. C., P. C. Hanson, R. Q. Thomas, A. B. Gerling, A. Hounshell, **A. S. L. Lewis**, M. E. Lofton, R. P. McClure, H. L. Wander, W. M. Woelmer, B. R. Niederlehner, M. E. Schreiber. 2022. Anoxia decreases the magnitude of the carbon, nitrogen, and phosphorus sink in freshwaters. *Global Change Biology*, 28(16), 4861–4881. <https://doi.org/10.1111/gcb.16228>
- 2021 Carey, C. C., W. M. Woelmer, M. E. Lofton, R. J. Figueiredo, B. J. Bookout, R. S. Corrigan, V. Daneshmand, A. G. Hounshell, D. W. Howard, **A. S. L. Lewis**, R. P. McClure, H. L. Wander, N. K. Ward, R. Q. Thomas. 2021. Advancing lake and reservoir water quality management with near-term, iterative ecological forecasting. *Inland Waters*, 12(1), 107–120. <https://doi.org/10.1080/20442041.2020.1816421>

- Woelmer, W. M., L. M. Bradley, L. T. Haber, D. H. Klimes, **A. S. L. Lewis**, E. Mohr, C. L. Torrens, K. I. Wheeler, A. M. Willson. 2021. 10 Simple Rules for training yourself in an emerging field. *PLOS Computational Biology*, 17(10), e1009440. <https://doi.org/10.1371/journal.pcbi.1009440>
- 2020 **Lewis, A. S. L.**, B. S. Kim, H. L. Edwards, H. L. Wander, C. M. Garfield, H. E. Murphy, N. D. Poulin, S. D. Princiotta, K. C. Rose, A. E. Taylor, K. C. Weathers, C. R. Wigdahl-Perry, K. Yokota, D. C. Richardson, D. A. Bruesewitz. 2020. Prevalence of nitrogen and phosphorus colimitation of freshwater phytoplankton explained by nitrogen deposition and lake characteristics across northeastern United States. *Inland waters*, 10(1), 42–50. <https://doi.org/10.1080/20442041.2019.1664233>
- 2018 Frassl, M. A, D. P. Hamilton, B. A. Denfeld, E. de Eyto, S. E. Hampton, P. S. Keller, S. Sharma, **A. S. L. Lewis**, G. A. Weyhenmeyer, C. M. O'Reilly, M. E. Lofton, N. Catalán. 2018. Ten simple rules for collaboratively writing a multi-authored paper. *PLOS Computational Biology*, 14(11), e1006508. <https://doi.org/10.1371/journal.pcbi.1006508>
- 2017 Lewis, S. E., J. J. Piatt., **A. S. L. Lewis**. 2017. Impact of a diet of native or non-native leaves on an amphipod *Gammarus pseudolimnaeus*. *Freshwater Science*, 36(4), 739–749. <https://doi.org/10.1086/694855>

OTHER TECHNICAL WRITING

- 2025 Peters, J., **A. S. L. Lewis**, A. Willson, C. Kowalski, C. Brookson, G. Koren, H. Moustahfid, H. O'Grady, J. McLachlan, J. Zobitz, M. Lofton, R. Krasnow, S. Shafiei Sabet. Resources for Reviewing Code. Ecological Forecasting Initiative Blog. <https://ecoforecast.org/resources-for-reviewing-code/>
- 2024 American Meteorological Society. 2024. The Future of Ecological Forecasting.
 • Contributed to conceptualization, drafting, and revision
- 2023 Bickley, S. L., S. Sickler, **A. S. L. Lewis**, C. D. Davis, K. Macdonald. Weiss reservoir is a source of greenhouse gas emissions. *Public comment submitted to the U. S. Federal Energy Regulatory Commission*. https://elibrary.ferc.gov/eLibrary/filelist?accession_num=20230113-5184
- 2021 **Lewis, A. S. L.**, B. Toh, J. Zwart, A. Shiklomanov, L. Johnson, E. White, H. Moustahfid, K. Heilman, A. Griffin, J. Peters, Q. Thomas, M. Dietze. 2021. Uncertainty quantification, Data assimilation, Modeling & Statistics. *Ecological Forecasting Initiative Task View*. <https://projects.ecoforecast.org/taskviews/uncertainty-quantification-data-assimilation-modeling-statistics.html>

NOVEL CODE PUBLICATIONS

- 2025 **Lewis, A. S. L.** and D. C. Richardson 2025. abbylouis/Seasons_in_lakes: Initial code release for manuscript submission. *Zenodo*. <https://doi.org/10.5281/zenodo.15786637>

- Lewis, A. S. L.** 2025. abbyLewis/SMARTX_CH4cast: Forecasts of methane (CH₄) fluxes from the Salt Marsh Accretion Response to Temperature eXperiment (SMARTX) (v1.0.1). *Zenodo*. <https://doi.org/10.5281/ZENODO.15352913>
- Lewis, A. S. L.** 2025. abbyLewis/Spring_memory: Data analysis of bottom-water temperature and oxygen dynamics in 615 lakes (v1.1.0). *Zenodo*. <https://doi.org/10.5281/zenodo.10714287>.
- 2024 **Lewis, A. S. L.**, M. E. Lofton, A. Breef-Pilz, and F. Olsson. 2024. abbyLewis/BVR_Drawdown: Effects of a 2022 drawdown on water quality in Beaverdam Reservoir (v1.1.0) [R]. *Zenodo*. <https://doi.org/10.5281/zenodo.8330109>
- Howard, D., Brentrup, J., Richardson, D., **Lewis, A. S. L.**, Olsson, F., & Carey, C. (2024, January 27). dwh77/FCR_Metab: Metabolism estimates in Falling Creek Reservoir. *Zenodo*. <https://doi.org/10.5281/zenodo.10575646>
- 2023 **Lewis, A. S. L.** and M. P. Lau. 2023. abbyLewis/Anoxia-Begets-Anoxia: Data analysis of biogeochemical dynamics in 656 lakes (v1.1.1) [R]. *Zenodo*. <https://doi.org/10.5281/zenodo.10086950>
- Lewis, A. S. L.** 2023. abbyLewis/FeDOC: Effects of hypoxia on coupled carbon and iron cycling in two freshwater reservoirs (v1.1.0) [R]. *Zenodo*. <https://doi.org/10.5281/zenodo.7527419>
- Wander, H. L., **A. S. L. Lewis**, D. W. Howard, M. E. Lofton, W. M. Woelmer, B. L. Brown, and C. C. Carey. 2023. hlwander/bvr_zoops_code: Zooplankton exhibit multiple diel migration strategies and substantial interannual changes in community structure in a eutrophic reservoir: Code (v1.0) [R]. *Zenodo*. <https://doi.org/10.5281/zenodo.8417404>

DATA PUBLICATIONS (* indicates mentored undergraduate student).

For annually-revised datasets, the most recent citation is presented and the years of previous co-authored revisions are listed to the left.

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- 2025 Holmquist, J., E. L. Koontz, A. Carr, **A. S. L. Lewis**, P. Megonigal, G. Noyce, S. Parker, A. E. Stearns, S. H. Wulfin, C. M. Young, B. Roberts. 2025. NASA Blue Methane Field Monitoring for Louisiana and the Chesapeake Bay, 2019-2021. Figshare. <https://doi.org/10.25573/serc.28373957.v1>
- 2025 Carey C. C., A. Breef-Pilz, **A. S. L. Lewis**, K. K. Hoffman. 2025. Time
2024 series of high-frequency profiles of depth, temperature, dissolved
2023 oxygen, conductivity, specific conductance, chlorophyll a, turbidity, pH,
2022 oxidation-reduction potential, photosynthetically active radiation,
2021 colored dissolved organic matter, phycocyanin, phycoerythrin, and
2019 descent rate for Beaverdam Reservoir, Carvins Cove Reservoir, Falling
Creek Reservoir, Gatewood Reservoir, and Spring Hollow Reservoir in
southwestern Virginia, USA 2013-2024. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/75d1c9a57a49030c468ab19db3643c36>

- 2025 Carey, C.C., A. Breef-Pilz, and **A. S. L. Lewis**. 2025. Time series of
2024 methane and carbon dioxide diffusive fluxes using an Ultraportable
2023 Greenhouse Gas Analyzer (UGGA) for Falling Creek Reservoir and
2022 Beaverdam Reservoir in southwestern Virginia, USA during 2018-2024
ver 4. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/cdd88943c7990f28daa9b8ab7a6b9679>
- 2024 Carey, C. C., F. E. Olsson, B. R. Niederlehner, A. Breef-Pilz, and **A. S. L.**
2023 **Lewis**. 2024. Time series of dissolved methane and carbon dioxide
2022 concentrations for Falling Creek Reservoir and Beaverdam Reservoir in
2021 southwestern Virginia, USA during 2015–2023 ver 8. *Environmental
Data Initiative*.
<https://doi.org/10.6073/pasta/5a45f25ba0440a5efd32177a9103fb5f>
- 2024 Carey, C.C., A. Breef-Pilz, A. D. Delany, A. G. Hounshell, **A. S. L. Lewis**, H.
L. Wander, G. Haynie, M. Kricheldorf, and E. Tipper. 2024. Time series
of high-frequency sensor data measuring water temperature, dissolved
oxygen, conductivity, specific conductance, total dissolved solids,
chlorophyll a, phycocyanin, fluorescent dissolved organic matter,
turbidity at discrete depths, and water level in Beaverdam Reservoir,
Virginia, USA in 2009-2023 ver 4. Environmental Data Initiative.
<https://doi.org/10.6073/pasta/31bb6047e0ac367c60a61884338799c4>
- McAfee, B.J., M.E. Lofton, A. Breef-Pilz, K.J. Goodman, R.T. Hensley, K.K.
Hoffman, D.W. Howard, **A. S. L. Lewis**, D.M. McKnight, I.A. Oleksy,
H.L. Wander, C.C. Carey, A. Karpatne, and P.C. Hanson.
2024. LakeBeD-US: Ecology Edition - a benchmark dataset of lake
water quality time series and vertical profiles. Environmental Data
Initiative.
<https://doi.org/10.6073/pasta/c56a204a65483790f6277de4896d7140>
- 2023 **Lewis, A. S. L.**, B.R. Niederlehner, A. Das*, H.L. Wander, M.E. Schreiber,
and C.C. Carey. 2023. Experimental microcosm incubations assessing
the effect of hypoxia on aqueous iron and organic carbon, pH, sediment
organic carbon, and sediment iron-bound organic carbon ver 1.
Environmental Data Initiative.
<https://doi.org/10.6073/pasta/60a7784acef3038d3c8a16776a5b5746>
- Lewis, A. S. L.**, M. P. Lau, S. F. Jane, Y. Beeri-Shlevin, S. H. Burnet, F.
Clayer, H. Feuchtmayr, H. Grossart, D. W. Howard, H. Mariash, J.
Delgado-Martin, R. L. North, I. Oleksy, R. M. Pilla, K. C. Rose, A. P.
Smagula, R. Sommaruga, S. E. Steiner, P. Verburg, D. Wain, G. A.
Weyhenmeyer, and C. C. Carey. 2023. Dissolved oxygen, temperature,
chlorophyll-a, total phosphorus, total nitrogen, and dissolved organic
carbon at multiple depths in 822 lakes from 1921-2022 ver 1.
Environmental Data Initiative.
<https://doi.org/10.6073/pasta/2cd6628a942de2a8b12d2b19962712a0>
- Lewis, A. S. L.**, M. E. Schreiber, B. R. Niederlehner, A. Das*, and C. C.
Carey. 2023. Total organic carbon, total nitrogen, and iron-bound
organic carbon in surficial sediment and settling particulate material
from Falling Creek and Beaverdam Reservoirs in 2019 and 2021 ver 1.
Environmental Data Initiative.
<https://doi.org/10.6073/pasta/a1d49c266b57465daa863cde4b1d4b4e>

- Schreiber, M. E., C. E. Wood, **A. S. L. Lewis**, N. W. Hammond, K. M. Krueger, M. E. Lofton, R. P. McClure, Z. W. Munger, A. Breef-Pilz, N. K. Ward, A. B. Gerling, B. R. Niederlehner, A. G. Hounshell, M. F. Verne, and C. C. Carey. 2023. Sediment trap time series data for Beaverdam Reservoir and Falling Creek Reservoir in southwestern Virginia, USA 2018 through 2022 ver 1. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/3e5d11ce1a38542d95250cdce18d6987>
- 2022 Carey, C. C., H. L. Wander, R. P. McClure, M. E. Lofton, K. D. Hamre, J. P.
2021 Doubek, A. B. Gerling, **A. S. L. Lewis**, A. Breef-Pilz. 2022. Secchi depth data and discrete depth profiles of photosynthetically active radiation, temperature, dissolved oxygen, and pH for Beaverdam Reservoir, Carvins Cove Reservoir, Falling Creek Reservoir, Gatewood Reservoir, and Spring Hollow Reservoir in southwestern Virginia, USA 2013–2021 ver 10. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/887d8ab8c57fb8fdf3582507f3223cd6>
- 2022 Carey, C. C., **A. S. L. Lewis**, D. W. Howard, W. M. Woelmer, P. A. Gantzer, K. A. Bierlein, J. C. Little, WVWA. 2022. Bathymetry and watershed area for Falling Creek Reservoir, Beaverdam Reservoir, and Carvins Cove Reservoir ver 1. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/352735344150f7e77d2bc18b69a22412>
- Jane, S. F., J. L. Mincer, M. P. Lau, **A. S. L. Lewis**, J. T. Stetler, K. C. Rose. 2022. Temperature and dissolved oxygen profiles for three Swiss lakes: 1972–2016 ver 6. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/7c08d0a4417f73d733d2a7eba43f57ea>
- 2021 Carey, C.C., W. M. Woelmer, **A. S. L. Lewis**, A. Breef-Pilz, D. W. Howard,
2020 and B. J. Bookout. 2021. Time series of high-frequency sensor data measuring water temperature, dissolved oxygen, pressure, conductivity, specific conductance, total dissolved solids, chlorophyll a, phycocyanin, and fluorescent dissolved organic matter at discrete depths in Falling Creek Reservoir, Virginia, USA in 2018–2020 ver 5. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/88896f4a7208c9b7bddcf498258edf78>
- 2021 **Lewis, A. S. L.**, W. M. Woelmer, H. L. Wander, D. W. Howard, J. W. Smith, R. P. McClure, M. E. Lofton, N. W. Hammond, R. S. Corrigan, R. Q. Thomas, and C. C. Carey. 2021. Systematic review of near-term ecological forecasting literature published between 1932 and 2020 ver 1. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/c4bea94f100f39a6b73c7b9a577df214>
- Lofton, M.E., D.W. Howard, R.P. McClure, H.L. Wander, W.M. Woelmer, A.G. Hounshell, **A. S. L. Lewis**, and C.C. Carey. 2021. Time series of phytoplankton biovolume at the depth of the vertical chlorophyll maximum in Falling Creek Reservoir, Vinton, VA, USA 2016–2019 ver 1. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/2de760e8b72e474c31e42526f5360f9a>

- 2020 Wander, H. L., **A. S. L. Lewis**, H. L. Edwards, and D. C. Richardson. 2020. Zooplankton density and size data in Lake Awosting, Lake Minnewaska, and Mohonk Lake, NY, USA 2013–2018 ver 1. *Environmental Data Initiative*.
<https://doi.org/10.6073/pasta/befde8268750ff108b59d8198eb989a1>

TEACHING EXPERIENCE

- Summer 2025* **Instructor:** Data Science Skills Series
 Smithsonian Environmental Research Center (SERC)
- Developed and co-led a series of six data science training workshops for the SERC community, with topics including Git/GitHub, spatial data analysis, and advanced data visualization
- Fall 2023* **Co-instructor of record:** Advanced R Programming
 Virginia Tech
- Designed and taught a graduate seminar on advanced programming in R (13 students)
 - All teaching materials are available in an open-source [GitHub repository](#)
- Spring 2022* **Teaching Assistant:** Principles of Biology Laboratory
 Virginia Tech
- Taught three lab sections with 7–24 students each
- Fall 2018* **Teaching Assistant:** Ecology for Non-Majors
 Pomona College
- Coordinated laboratory exercises and field logistics for an introductory ecology class
- Spring 2018* **Writing Mentor:** Introduction to Ecology and Evolution
 Pomona College
- Led individualized mentor sessions to help undergraduate students develop skills in scientific writing
- Spring 2018* **Teaching Assistant:** Bilingual (Spanish and English) kindergarten
 Mountain View Elementary School
- Spring 2018* **Mentor:** Rooftop Garden Project
 The Draper Center for Community Partnerships, Pomona College
- Developed programming for biweekly workshops on earth science and environmental justice for local high school students

GUEST LECTURES

- 2025 **Data Carpentries**, SERC
- 2024 **Aquatic Ecology**, Carroll University
Introduction to Graduate Studies, Virginia Tech
- 2023 **Careers in Biological Sciences**, Virginia Tech
Ecology, Virginia Tech

UNDERGRADUATE STUDENTS MENTORED

- 2025 **Rania Adouim** (Amherst College)
- SERC REU intern. First-author on a presentation at the Ecological Society of America 2025 meeting and a manuscript in preparation
- 2023 **Ryan Keverline** (Virginia Tech)
- 2019–2023 **Arpita Das** (Virginia Tech)
- Coauthor on seven datasets, three presentations, and a published manuscript
- 2020 **Nick Ruszkowski** (Virginia Tech)

INVITED PRESENTATIONS (presenters underlined)

- 2025 **Lewis, A. S. L.** 2025. Leveraging marsh restoration for empowering climate change communication in northeastern National Parks. Invited talk at: *ESA Annual Meeting*; Baltimore, MD, USA.
- 2024 **Lewis, A. S. L.** and C. C. Carey. 2024. Synthesis of long-term water quality data from 615 lakes highlights seasonally-distinct effects of climate change on lake ecosystem function. Invited talk at: *ESA Annual Meeting*; Long Beach, CA, USA.
- Lewis, A. S. L.** 2024. Trailblazing Award Presentation. Invited talk at: *ESA Annual Meeting*; Long Beach, CA, USA.
- Lewis, A. S. L.** 2024. Oxygen dynamics in the bottom waters of lakes: Understanding the past to predict the future. Invited seminar at: *Helmholtz Centre for Environmental Research – UFZ*; Magdeburg, Germany.
- Lewis, A. S. L.** 2024. Deoxygenation of temperate lakes: Understanding the past to predict the future. Invited seminar at: *Lake Superior State University*; Sault Ste. Marie, MI, USA.
- 2023 **Lewis, A. S. L.** 2023. Using forecasts to understand ecosystems. Invited talk at: *American Meteorological Society 103rd Annual Meeting*; Boulder, CO, USA (virtual).
- Lewis, A. S. L.** 2023. Using forecasts to understand ecosystems. Invited talk at: *Ecological Forecasting Initiative Student Association Meeting* (virtual).
- 2022 **Lewis, A. S. L.**, W. M. Woelmer, H. L. Wander, D. W. Howard, J. W. Smith, R. P. McClure, M. E. Lofton, N. W. Hammond, R. S. Corrigan, R. Q. Thomas, C. C. Carey. 2022. Near-term ecological forecasting: state of the field. Invited talk at: *INTECOL 2022*; Geneva, Switzerland (virtual).
- Lewis, A. S. L.**, A. Hensley, C. López Lloreda, C. Perez, E. Bueren, G. Palissery, G. O'Malley, H. Wander, and S. Drew. 2022. Flip the Fair: an intersection of science communication, outreach, and empowerment. Invited talk at: *Virginia Tech Life Sciences Seminar*; Blacksburg, VA, USA.

- 2021 **Lewis, A. S. L.** 2021. *Virginia Tech graduate commencement address*. Blacksburg, VA, USA.
- Lewis, A. S. L.** 2021. Oxygen: stories of fire, beer, and the guillotine. *Science on Tap*; Blacksburg, VA, USA.
- Kroehler, C., P. Raun, J. Appiah-Kubi, C. Colleary, V. Diaz, **A. S. L. Lewis**, and A. Wendler. 2021. Girls Launch! Providing Female Scientist Role Models to Early Elementary Children. *Virginia Tech Widening Inclusivity in the (Geo)Sciences Seminar*; Blacksburg, VA, USA.
- 2020 **Lewis, A. S. L.** 2020. The Unseen Nature of Lakes. *Skype a Scientist LIVE*. Virtual Q&A.
- An invited 45-minute live-streamed Q&A with a general audience talking about lakes

SELECTED CONTRIBUTED PRESENTATIONS (presenters underlined; *indicates mentored undergraduate student)

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- 2025 **Adouim, R.***, **A. S. L. Lewis**, J. Wolfe, R. Cheney, J. Holmquist. Ten years of plant surveys reveal interactive effects of climate change and biological invasion on community composition in a coastal marsh. Poster presentation at: *ESA Annual Meeting*; August 2025; Baltimore, MD.
- Lewis, A. S. L.**, J. R. Holmquist, A. N. Al-Haj, R. L. Rich, and G. L. Noyce. Warming has non-linear effects on the predictability of methane fluxes in a brackish wetland. Oral presentation at: *Ecological Forecasting Initiative Meeting*; May 2025; Blacksburg, VA.
- 2024 **Lewis, A. S. L.**, J. Rabaey, K. Attermeyer, P. Aurich, S. Bansal, B. Bertolet, R. Bhattacharya, I. Bussmann, S. B. Cadieux, E. Calamita, C. Carey, F. Clayer, T. Davidson, B. R. Deemer, B. Denfeld, W. Eckert, C. Esposito, A. Gorsky, N. Griffiths, H-P. Grossart, D. Hamilton, M. Holgerson, B. Huser, T. Iwata, J. Jansen, S. Juutinen, P. Kortelainen, M. Koschorreck, T. Kragh, A. Laas, T. Larmola, I. Laurion, M. F. Lehmann, L. Liu, A. Matoušů, D. Nizzoli, C. Ordóñez, M. Peacock, R. Pilla, J. Pu, T. Riis, T. Saarela, A. B. Santoso, C. Schubert, B. S. Sherman, J. S. Sør, K. Strock, K. Stenehjem, K. Tsuchiya, K. Wendt-Potthoff, G. A. Weyhenmeyer, and P. Znachor. Patterns and drivers of bottom-water greenhouse gas concentrations across 522 lakes worldwide. Oral presentation at: *AGU Annual Meeting*; December, 2024; Washington, D.C.
- Lewis, A. S. L.** and C. C. Carey. Ecological memory of spring air temperature drives summer water quality dynamics in temperate lakes. Oral presentation at: *ASLO Meeting*; June 2024; Madison, WI.
- 2023 **Lewis, A. S. L.**, W. Zhi, J. Mai, N. Basu, C. C. Carey. Ecological memory of climate and land use change shapes water quality across >650 widespread lakes. Oral presentation at: *AGU Annual Meeting*; December 2023. San Francisco, CA.
- Outstanding Student Presentation award

Lewis, A. S. L., M. Lau, S. Jane, K. Rose, Y. Be'eri-Shlevin, S. H. Burnet, F. Clayer, O. Erina, H. Feuchtmayr, H. P. Grossart, T. Harris, D. Howard, A. James, H. Mariash, J. Delgado Martin, R. North, I. Oleksy, R. M. Pilla, J. A. Rusak, R. Sommaruga, P. Verburg, D. Wain, J. Watkins, G. Weyhenmeyer, C. C. Carey. Anoxia Begets Anoxia: a positive feedback to the deoxygenation of temperate lakes. Poster presented at: *GLEON All-Hands Meeting*; June 2023; Ryn, Poland.

2022

Lewis, A. S. L., M. Lau, S. Jane, K. Rose, Y. Be'eri-Shlevin, S. H. Burnet, F. Clayer, O. Erina, H. Feuchtmayr, H. P. Grossart, T. Harris, D. Howard, A. James, H. Mariash, J. Delgado Martin, R. North, I. Oleksy, R. M. Pilla, J. A. Rusak, R. Sommaruga, P. Verburg, D. Wain, J. Watkins, G. Weyhenmeyer, C. C. Carey. Declining oxygen concentrations are associated with increased phosphorus and chlorophyll-a across 504 lakes. Poster presented at: *GLEON All-Hands Meeting*; November 2022; Lake George, NY.

Lewis, A. S. L., W. M. Woelmer, H. L. Wander, D. W. Howard, J. W. Smith, R. P. McClure, M. E. Lofton, N. W. Hammond, R. S. Corrigan, R. Q. Thomas, C. C. Carey. Near-term ecological forecasting: state of the field. Oral presentation at: *Ecological Forecasting Initiative Meeting*; May 2022; virtual.

Lewis, A. S. L., B. R. Niederlehner, A. Das*, N. W. Hammond, M. E. Schreiber, C. C. Carey. Effects of Hypoxia on Coupled Iron and Carbon Cycling Differ by Timescale in Two Freshwater Reservoirs. Oral presentation at: *Interfaces of Global Change Research Symposium*; May 2022; Blacksburg, VA.

- 1st Place Platform Presentation and Karen P. DePauw Outstanding Interdisciplinary Presentation Award

Lewis, A. S. L., B. R. Niederlehner, A. Das*, N. W. Hammond, M. E. Schreiber, C. C. Carey. Effects of Hypoxia on Coupled Iron and Carbon Cycling Differ by Timescale in Two Freshwater Reservoirs. Oral presentation at: *Joint Aquatic Sciences Meeting*; May 2022; Grand Rapids, MI.

Lewis, A. S. L., C. López Lloreda, G. O'Malley, H. Wander, E. Bueren, S. Drew, A. Hensley, C. J. Kroehler, A. D. Lowery, G. K. Palissery, C. Perez, C. Pihlstrom, P. Raun. Flip the Fair: an intersection of science communication, outreach, and empowerment. Oral presentation at: *Joint Aquatic Sciences Meeting*; May 2022; Grand Rapids, MI.

2021

Lewis, A. S. L., B. R. Niederlehner, A. Das*, N. W. Hammond, M. E. Schreiber, C. C. Carey. High levels of iron-bound organic carbon correspond to multi-annual oxygenation history in two reservoirs. Poster presented at: *GLEON All-hands Meeting*; October 2021; virtual.

Lewis, A. S. L., R. P. McClure, P. C. Hanson, C. C. Carey. Integrated whole-ecosystem experiments and ecosystem modeling reveal that seasonal oxygen depletion is driven primarily by temperature in a eutrophic reservoir. Oral presentation at: *ASLO Meeting*; June 2021; virtual.

- 2020 **Lewis, A. S. L.**, M. E. Lofton, R. P. McClure, W. M. Woelmer, P. C. Hanson, R. Q. Thomas, C. C. Carey. Near-term, iterative forecasts highlight the relative importance of two drivers for dynamic oxygen concentrations in a drinking water reservoir. Oral presentation at: *AGU Annual Meeting*; December 2020; virtual.
- Lewis, A. S. L.**, M. E. Lofton, R. P. McClure, W. M. Woelmer, P. C. Hanson, R. Q. Thomas, C. C. Carey. Bottom-water temperatures drive changing rates of oxygen depletion in a drinking water reservoir. Poster presented at: *GLEON All-hands Meeting*; October 2020; virtual.
- Lewis, A. S. L.**, M. E. Lofton, R. P. McClure, W. M. Woelmer, P. C. Hanson, R. Q. Thomas, C. C. Carey. Near-term, iterative ecological forecasts provide insight into the drivers of changing oxygen concentrations in a drinking water reservoir. Poster presented at: *ESA Annual Meeting*; August 2020; virtual.
- Carey, C. C., **Lewis, A. S. L.**, Hounshell, A. G., Howard, D. W., McClure, R. P., Hammond, N. W., Lofton, M. E., P. C. Hanson, J. C. Little, M. Schreiber, F. Birgand. Dynamic carbon-oxygen interactions over minute to annual time scales in an experimentally-oxygenated reservoir. Oral presentation at: *EGU General Assembly*; May 2020; virtual.
- Lewis, A. S. L.**, P. C. Hanson, C. C. Carey. Oxygen demand can be predicted by oxygenation rates and temperature. Oral presentation at: *Virginia Water Conference*; March 2020; Richmond, VA.
- 2019 **Lewis, A. S. L.**, A. G. Hounshell, R. P. McClure, N. W. Hammond, D. W. Howard, M. E. Lofton, H. L. Wander, W. M. Woelmer, F. Birgand, P. C. Hanson, J. C. Little, M. E. Schreiber, C. C. Carey. Consequences of changing oxygen availability for carbon cycling in freshwater ecosystems. Poster presented at: *GLEON All-hands Meeting*; November 2019; Huntsville, ON.
- 2017 **Lewis, A. S. L.**, D. C. Richardson, K. Weathers. Depth Affects Nutrient Limitation of Phytoplankton within a Stratified Lake. Poster presented at: *Council on Undergraduate Research REU Symposium*; October 2017; Alexandria, VA.
- By nomination from the Cary Institute for Ecosystem Studies
- Kim, B. S., **A. S. L. Lewis**, H. L. Edwards, H. L. Wanders, A. E. Taylor, N. Poulin, S. Princiotta, K. Yokota, C. Wigdahl, K. Rose, D. C. Richardson, D. A. Bruesewitz. Patterns of Nutrient Limitation in Sixteen Northeastern United States Lakes. Poster presented at: *GLEON All-hands Meeting*; October 2017; New Paltz, NY.

SELECTED OUTREACH

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- 2024–
2025 **Science Communication in the Parks Fellow**
- Developed digital and physical materials for two national parks to communicate the value of coastal marsh restoration (\$3700 stipend)

- 2019–
2024 **Outreach Committee Lead**, Virginia Tech Stream Team
- Organized a “flipped science fair” where elementary students judged 27 graduate student posters
 - Coordinated summer camp activities at the Blacksburg Nature Center
 - Built, painted, and installed a “stream box” for kids to learn about streams
 - Coordinated Wikipedia editing sessions to highlight women and minoritized researchers
- 2021–
2024 **Virginia Scientist-Community Interface (VS-CI)**
Reservoir Greenhouse Gas Emissions Team Member
- Used the G-RES tool to quantify greenhouse gas emissions from five reservoirs in Alabama with non-profit partners
- 2022–
2023 **Letters to a Pre-Scientist Pen-pal**
- Exchanged letters with a 6th grade student to broaden perceptions of STEM
- 2020–
present **Science Fair Judge**
- Virginia State Science and Engineering Fair: 2020, 2021
 - Blue Ridge Highlands Regional Science Fair: 2020–2024
 - Terra Rochester Finger Lakes Science & Engineering Fair: 2020
- 2020 **Girls Launch! Fellow**, Blacksburg, VA
- Developed an activity and [video](#) for kindergarten students to learn about water and see female scientists in action (\$1,000 stipend)
- 2018–
2020 **Envirobites (envirobites.org)**
Contributor and editor
- Published [16 articles](#) communicating environmental research to a general audience
- 2015–
2019 **Pomona College Sustainability Office**
- Initiated a Zero Waste Events Program to eliminate trash from major campus events (featured in The Week and on the United Nations website)
 - Collaborated on a published Action Plan with sustainability initiatives and goals for the next 15 years
 - Mentored interns and developed programming to promote sustainability

SELECTED PROFESSIONAL SERVICE

- 2026 **ASLO-SIL 2026 Joint Meeting**
- Co-organizer: “Gassy Waters: Advances in Understanding and Measuring Gas Fluxes across the Aquatic Continuum” (accepted)
- 2024–
present **American Meteorological Society**
- Member: Committee on Ecological Forecasting
- 2023–
present **American Geophysical Union**
- Co-organizer: “Ecological Forecasting in Managed and Natural Systems: Hydrology, Agriculture, and the Earth System” session (2025)
 - Lead organizer and Outstanding Student Presentation Award chair: “Model-Data Integration and Novel Paradigms in Ecosystem Forecasting” session (2024)
 - Co-organizer: “Ecological Forecasting in the Earth System” session (2023)

- 2020–
present **Ecological Forecasting Initiative**
- Co-chair: Theory Working Group (2020–present)
 - Member: Education Working Group (2022–present)
 - Member: Student and Early Career Association (2020–2024)
- 2021–
2025 **Global Lake Ecological Observatory Network**
- Co-moderator: Metabolism Working Group

PEER-REVIEW

Manuscript review for:

- *Global Change Biology*
- *Limnology and Oceanography Letters*
- *Limnology and Oceanography*
- *Journal of Geophysical Research – Biogeosciences*
- *Global Ecology and Biogeography*
- *Ecosphere*
- *Water Resources Research*
- *Scientific Data*
- *Inland Waters*
- *Freshwater Biology*

Grant review for:

- Deutsche Forschungsgemeinschaft (DFG; German Research Foundation)
- Graduate Research Development Program, Virginia Tech

SOCIETY AFFILIATIONS

- Global Lake Ecological Observatory Network (GLEON)
- Ecological Forecasting Initiative (EFI)
- Ecological Society of America (ESA)
- Association for the Sciences of Limnology and Oceanography (ASLO)
- American Geophysical Union (AGU)
- American Meteorological Society (AMS)
- Society for Open, Reliable, and Transparent Ecology and Evolutionary biology (SORTEE)